



Finetune Large Pre-Trained Model Based on Frequency-Wise Multi-Query Attention Pooling for Anomalous Sound Detection

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Abstract

Existing anomalous sound detection (ASD) methods generally rely on the learned patterns of normal sounds to identify deviations as anomalies in generative and discriminative ways. However, partly due to the diverse and sparse distribution of anomalies, they may suffer from poor generalization capability to distinguish unseen normal sounds and anomalies. In this paper, we propose to finetune the large pre-trained model based on frequency-wise multi-query attention pooling (FMQAP) for ASD. Specifically, FMQAP first dynamically fuses multiple output embeddings from the pre-trained model, and then exploits a novel frequency-wise attention pooling scheme to improve the generalization capability for various machine types. Evaluations on DCASE 2023 Task2 benchmark demonstrate the superiority of FMQAP, achieving the state-of-the-art 77.62% in terms of official score.

Index Terms: anomalous sound detection, pre-trained models, frequency-wise, attention pooling

1. Introduction

Anomalous sound detection (ASD) aims to automatically identify anomalous states through audio signal analysis during machine operation [1, 2]. Compared with conventional monitoring approaches, ASD enables non-contact, cost-effective, and efficient intelligent monitoring, with broad application potential in industrial manufacturing domains.

Due to the scarcity and diversity of anomalies, existing ASD methods typically train the model to characterize the distribution of normal audio, and samples deviate from it are identified as anomalies [3, 4]. For example, autoencoders (AE) [5, 6] are used to model the distribution of normal samples in an unsupervised way. However, due to the strong generalization ability of generative based methods, the performance is generally far from satisfactory. Discriminative based methods [7–10] utilize labels from meta annotation to train the convolutional neural networks (CNN). From the perspective of outlier exposure (OE) [11], they treat the normal data from other categories as outliers to learn a discriminative and compact distribution.

Extensions have been proposed by exploiting the time and frequency domain information. Guan et.al [12] proposed the Time Weighted Frequency Domain Representation with GMM (TWFR-GMM) ASD method, which can adapt to different machine types. Chen et.al. [13] proposed to use statics exchange (StEx) on time or frequency axis of input spectrogram to construct the pseudo anomalies. In [14], high-pass filtering was applied to the audio of certain machine. Li et al. [15] designed filterbanks with varying distribution densities across different frequencies for each machine type in the MIMII dataset [16].

These studies indicate that different machines exhibit distinct frequency patterns, emphasizing the significance of specific frequencies and selecting appropriate frequency features for ASD.

Despite the promising performance of the discriminative based ASD methods, they heavily rely on the labels extracted from meta information, which may not be able to address the first-shot problem in DCASE 2023 Challenge Task 2 [2]. The challenge is to train model for a complete new machine, where machine types in the development and evaluation datasets are completely different. This may incur several issues that significantly degrade the ASD performance. 1) The model learned from the normal data in training set can hardly discriminate whether the deviations are from normal data of new machine types or from anomalies. 2) The domain shifts between source and target domains of development dataset caused by various factors, including different machine's operation conditions and environment, differ from those of evaluation. This issue cannot be resolved solely by using additional training dataset or data augmentation techniques. It is a dilemma to balance the precision and recall metrics that are related to the miss- and false-alarm in detection systems.

Several works have been proposed for first-shot problem [15, 17, 18]. For example, Han et al. [17] suggested to leverage self-supervised pre-trained models on large large-scale speech data. By fusing the output of different layers and data augmentation to simulate different machine operating sets. Jiang et al. [18] finetuned the pre-trained models on AudioSet [19] to adapt to various conditions of machine audio. Leveraging large pre-trained models enables the acquisition of common knowledge about speech and audio from extensive training data, effectively mitigating the issue of data scarcity [18].

However, existing methods commonly focus on reduce the inconsistency of the pre-trained model and inductive bias of machine audio. BEATs and ATST-frame use average-pooling to aggregate the patch-level features into a segment-level representation [20, 21]. For ASD, Anopatch [18] and Conformer [11] employ attentive statistics pooling (ASP) [22]. The simple averaging or weighted summing aggregation scheme disregards the time-frequency characteristics in audio signal, which has been demonstrated to be effective for ASD tasks [6, 12, 15, 23].

In this paper we propose a novel frequency-wise multi-query attention pooling (FMQAP) method to address the first-shot problem by exploiting the pre-trained model, as shown in Fig.1. The key idea is to exploit information extracted from multiple transformer layers, frequency sub-bands to improve the robustness and effectiveness of first-shot problem in ASD. Specifically, the pre-trained model on AudioSet is used as feature encoder to extract the patch-level features. Following [17], a layer-wise weighted average pooling (L-WAP) is designed to

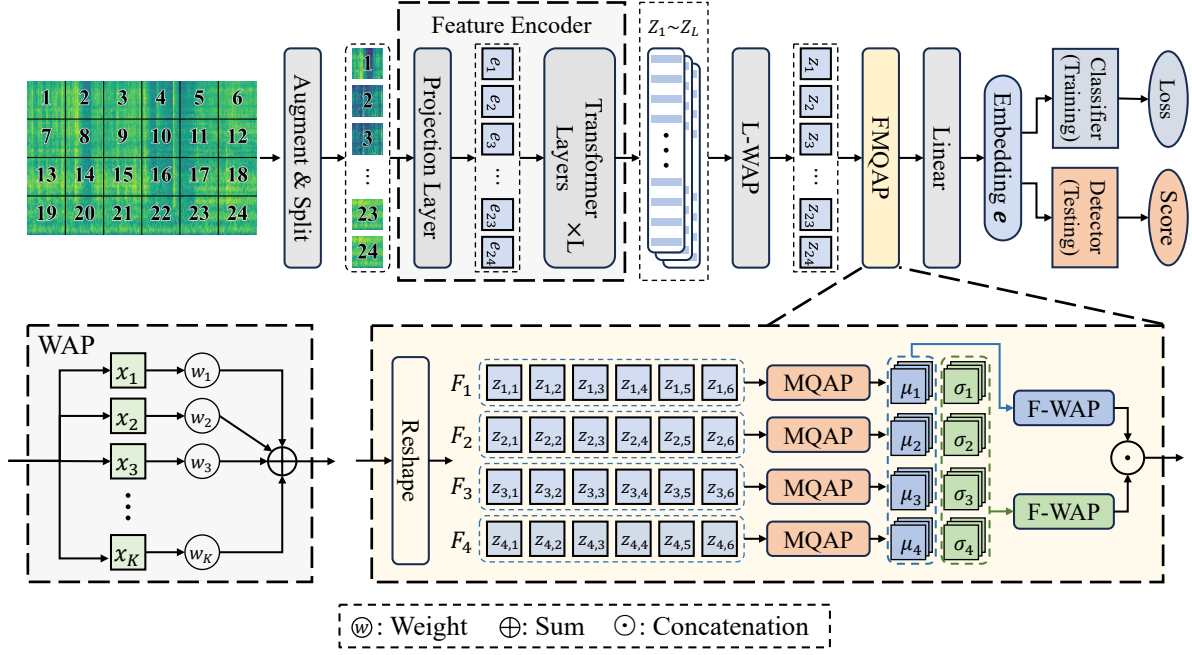


Figure 1: Layer-wise Transformer-based decomposition of non-overlapping multi-query patches utilizing learnable time and frequency attention weights. “WAP” denotes weighted average pooling.

aggregate the features from different transformer layers. After that, FMQAP is proposed to aggregate the patch-level features from frequency sub-bands, where a cross-attention mechanism with multiple queries is applied to improve ASD performance of first-shot problem. Finetuning on DCASE 2020 and DCASE 2023 Task 2 benchmarks has demonstrated the effectiveness of that our method, achieving 95.34% and 77.62% of mean score and official score, respectively, which significantly outperform the state-of-the-arts (SOTA).

2. Methodology

2.1. Overview of FMQAP based ASD method

Fig. 1 presents the overall framework of our method. A feature encoder, initialized with BEATs, serves as the feature extractor to obtain patch-level representations from the input mel spectrogram. The output from each Transformer layers is first dynamically fused using L-WAP. And then, we aggregate the fused features into a clip-level embedding e using the proposed FMQAP, which is used for finetuning ASD model.

Specifically, given an input audio, we first extract the mel-spectrogram $X \in \mathbb{R}^{F \times T}$. After random masking in both time and frequency dimensions as spec-augmentation [30], the spectrogram is then divided into non-overlapping patches $\{x_{f,t}\}$, forming a 2D grid:

$$X = \{x_{f,t}\}_{f=1, t=1}^{N_F, N_T}, \quad x_{f,t} \in \mathbb{R}^{F' \times T'} \quad (1)$$

where $N_F = \lfloor F/F' \rfloor$ and $N_T = \lfloor T/T' \rfloor$ denote the number of patches along the frequency and time dimensions, respectively.

Next, the patches are flattened into a sequence and fed into a vision Transformer (ViT) backbone initialized with BEATs. This process produces an output vector sequence for each transformer layer:

$$Z_l = \{z_{i,l}\}_{i=1}^M, \quad z_i \in \mathbb{R}^D \quad (2)$$

where $M = N_F \times N_T$ represents the total number of patches, and l indicates the layer index of the transformer layers.

Following [17], we employ the L-WAP to exploit the outputs from different Transformer layers, and obtain a more comprehensive feature representation by capturing complementary information. L-WAP computes a weighted sum of the outputs from each layer using a set of learnable parameters $\{w_l\}_{l=1}^L$, which enables the model to adaptively select informative layers according to finetuning.

$$Z = \sum_{l=1}^L w_l \cdot Z_l \quad (3)$$

Given the fused patch-level features Z , the proposed FMQAP is then applied to aggregate them into a clip-level representation e , detailed in section 2.2. A linear projection layer is then used to reduce e into lower-dimensional embedding, followed by a classifier. In the training phase, the ArcFace loss [31] L is utilized to optimize the model by predicting labels extracted from metadata, which is as follows.

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cdot (\cos(\theta_{y_i} + m))}}{e^{s \cdot (\cos(\theta_{y_i} + m))} + \sum_{j \neq y_i} e^{s \cdot \cos(\theta_j)}} \quad (4)$$

where θ_j is the angle between the embedding e and the weight vector corresponding to the j -th class, representing the angular closeness between the quantized sample and each class. The margin m and scaling factor s are hyperparameters, with default values of $m = 0.1$ and $s = 30$.

In the testing phase, the finetuned network is utilized to extract embeddings from the training set, and construct the distribution of normal samples. The detection is then performed by using a k-nearest neighbors (KNN) backend [32], which uses the distance between a test sample and the normal samples to compute the anomaly score.

Table 1: Results on DCASE 2020 Task 2 dataset for proposed and existing methods

Models	Development set							Evaluation set						
	fan	pump	slider	valve	ToyCar	ToyCon.	mean	fan	pump	slider	valve	ToyCar	ToyCon.	mean
Ranked 1st [24]	80.65	83.27	93.41	94.30	92.72	73.28	86.27	89.42	87.69	93.68	93.51	92.04	82.27	89.77
Ranked 2nd [25]	91.18	94.94	96.20	95.23	95.93	81.46	92.49	97.77	92.65	95.08	84.19	88.70	84.04	90.40
ASD-AFPA [26]	95.52	90.61	99.05	97.27	92.80	70.35	90.93	-	-	-	-	-	-	-
PAE+GeCo [27]	88.80	90.32	97.75	98.34	94.40	74.32	90.66	-	-	-	-	-	-	-
AnoPatch [18]	86.46	93.10	99.20	97.53	96.10	73.20	90.93	95.56	94.34	99.77	96.26	96.00	83.74	94.28
baseline	82.13	88.36	98.30	94.86	92.99	69.89	87.76	95.82	88.51	93.00	97.15	90.21	77.51	90.37
FMQAP	85.30	94.17	98.82	98.85	95.62	74.59	91.23	97.41	94.40	98.95	99.49	97.93	83.83	95.34

Table 2: Results on DCASE 2023 Task 2 dataset for proposed and existing methods

Models	Development set								Evaluation set							
	ToyCar	ToyTrain	bearing	fan	gearbox	slider	valve	hmean	ToyDrone	ToyNscale	ToyTank	Vacuum	bandsaw	grinder	shaker	hmean
Ranked 1st [28]	54.58	54.20	64.21	75.95	76.92	92.06	85.67	69.28	55.71	84.92	60.72	92.27	60.97	65.18	63.50	66.97
Ranked 2nd [29]	58.92	55.92	62.62	66.66	77.77	83.96	68.61	66.56	52.89	71.90	70.73	91.48	55.47	64.76	70.98	66.39
Han [17]	63.47	57.35	57.10	62.76	67.52	79.11	67.79	64.31	-	-	-	-	-	-	-	-
AnoPatch [18]	58.78	67.16	70.43	66.65	58.67	81.88	53.73	64.24	64.49	85.04	72.60	92.24	69.71	64.06	80.25	74.23
baseline	55.92	53.99	70.76	62.52	68.77	86.21	67.55	65.13	60.15	80.87	72.57	92.26	57.12	65.01	81.44	70.87
FMQAP	56.69	59.11	74.94	72.05	69.24	93.72	63.45	68.19	67.92	94.41	77.20	99.56	63.08	72.31	82.02	77.62

2.2. Frequency-Wise Multi-Query Attention Pooling

Most existing pooling methods are designed for frame-level feature representation, neglecting the crucial frequency characteristics of audio data. Different machines exhibit distinct frequency patterns. As shown in [12–15], selecting appropriate frequency sub-band may be beneficial for ASD, especially in case of first-shot problem. In [18], statistical pooling is used to extract the first and second order statistics according to the patch-level features. However, the important frequency information may be ignored. To address this, FMQAP is proposed to aggregate frequency-level features across different sub-bands by exploiting the cross sub-band context information.

Specifically, given the patch-level feature sequence $Z = \{z_i\}_{i=1}^M, z_i \in \mathbb{R}^D$, we reshape it into a frequency-wise representation following the 2D grid order:

$$Z' = \{z_{f,t}\} \in \mathbb{R}^{N_F \times N_T \times D} \quad (5)$$

For each sub-band, the output feature is represented as $Z'_f = \{z_{f,:}\} \in \mathbb{R}^{N_T \times D}$. The N learnable queries $Q_f = \{q_{n,f}\}_{n=1}^N \in \mathbb{R}^{N \times D}$ are used for the attention score computation along the time dimension following the self-attention mechanism in Transformer, which is as follows:

$$a_{t,n,f} = \frac{\exp(q_{n,f} \cdot z_{f,t}^T)}{\sum_{\tau=1}^{N_T} \exp(q_{n,f} \cdot z_{f,\tau}^T)} \quad (6)$$

where the attention score $a_{t,n,f}$ indicates the importance of the query $q_{n,f}$ to each temporal patch within the specified frequency sub-band f . These attention scores are then utilized to compute frequency-wise weighted statistics as follows.

$$\mu_{n,f} = \sum_{t=1}^{N_T} a_{t,n,f} \cdot z_{f,t} \quad (7)$$

$$\sigma_{n,f} = \sqrt{\sum_{t=1}^{N_T} a_{t,n,f} \cdot z_{f,t}^2 - \mu_{n,f}^2} \quad (8)$$

The statistics μ and σ of attentively weighted features of all sub-bands. After L_2 normalization, the clip-level representation

can be obtained by concatenating μ and σ .

$$\mu = [\mu_{n,f}]_{n=1,f=1}^{N,N_F}, \quad \sigma = [\sigma_{n,f}]_{n=1,f=1}^{N,N_F} \quad (9)$$

$$e = [\mu, \sigma] \in \mathbb{R}^{D \times 2 \times N \times N_F} \quad (10)$$

where $[\cdot]$ denotes concatenation. However, the high dimension $D \times 2 \times N \times N_F$ of e may lead to an excessive number of parameters in the subsequent linear layer used for dimension reduction. One potential solution is to reduce the query number N to improve the model efficiency with the cost of degrading performance. To address this issue, we introduce the WAP again, termed frequency-wise weighted average pooling (F-WAP), with two sets of learnable parameters, $\{\alpha_{n,f}\}$ and $\{\beta_{n,f}\}$, to fuse e as follows.

In experiment section, we demonstrate that F-WAP can better trade-off between the complexity and performance.

3. Experiments and Results

3.1. Datasets

We evaluate our method on the DCASE 2020 Task 2 and DCASE 2023 Task 2 datasets. The DCASE 2020 dataset comprises parts of ToyADMOS [33] and the MIMII [16] dataset, which contains 6 machine types with multiple machine IDs per type. Each ID includes approximately 1,000 normal audio clips in the training set and 400 mixed normal/anomalous clips in the evaluation set. The DCASE 2023 dataset comprises parts of ToyADMOS2 [34] and the MIMII DG [35] dataset, which extends the challenge with domain shift scenarios across 14 machine types. Each machine type contains a single section composed of 990 source-domain samples and 10 target-domain samples. The evaluation set provides 200 test samples per machine section, reflecting diverse operational states through combinations of machine type, section, and attributes.

3.2. Evaluation Metrics

Following DCASE challenge protocols [1, 2], we adopt the area under the receiver operating characteristics curve (AUC) and partial AUC (pAUC) with a low false positive rate restriction (FPR) range $[0, 0.1]$ as metrics.

Table 3: Ablation Study of the Proposed Approach on the DCASE 2023 Task 2 evaluation dataset.

System	Params	Source AUC	Target AUC	pAUC	Official Score
baseline	91M	72.11	78.74	63.43	70.87
+ sepc-aug	91M	78.23	72.78	63.40	70.93
+ ArcFace	91M	82.51	75.58	67.11	74.53
+ FMQAP	166M	83.04	79.16	70.27	77.11
+ F-WAP	91M	83.61	80.01	69.51	77.23
+ L-WAP	91M	84.02	80.45	69.83	77.62

For DCASE 2020, we report arithmetic means of per-machine AUC (%) and pAUC scores (%). For DCASE 2023, we report harmonic means of source-domain AUC (SourceAUC, %), target-domain AUC (TargetAUC, %), and pAUC per machine type, followed by the overall harmonic mean across all machine types, aligning with official evaluation criteria.

3.3. Implementation Details

Audio inputs are resampled to 16 kHz and processed into 10-second clips. Mel spectrograms with 128 frequency bands are extracted using 25 ms windows and a hop size of 10 ms. These spectrograms are then split into patches with patch-size of 16×16 . The BEATs_{iter3} backbone [20] with 12 transformer layers (768-dim hidden states) is used as the feature extractor. Optimization is performed using the AdamW optimizer [36], with a batch size of 64, an peak learning rate of 1×10^{-4} , and a total of 100 training epochs are trained. The learning rate is warmed up for the first 10 epochs, linearly increasing to its target value, and then decayed following a cosine schedule to 5×10^{-8} over the remaining epochs. The ArcFace loss parameters are set with margin 0.1 and scale 30. In FMQAP, $N=24$ learnable queries are used, followed by $K = 1$ KNN-based anomaly scoring.

In the baseline ASD system, Cross-Entropy (CE) loss is used together with average pooling, without L-WAP or any data augmentation. All the other settings are kept unchanged.

3.4. Results

Tables 1 and 2 present our experimental results on the DCASE 2020 and DCASE 2023 datasets, respectively. We compare our method with previous SOTA methods, including those with pre-trained large model.

On DCASE 2020 development and evaluation datasets, our method achieves an average score of 91.23% and 95.34%, respectively, surpassing both our baseline system and previous SOTA approaches. On the DCASE 2023 development and evaluation datasets, our method obtains official scores of 68.19% and 77.62%, significantly outperforming our baseline system and prior SOTA methods. These results demonstrate the effectiveness of our proposed approach.

3.5. Ablation Study

To further explore the effectiveness of the proposed approach, we conducted a series of ablation experiments on the DCASE 2023 evaluation dataset, as shown in Table 3. The results indicate that incorporating ArcFace loss and our FMQAP method leads to significant performance improvements, with respective gains of 3.60% and 2.58%. In particular, FMQAP improves performance in the target domain and enhances detection un-

Table 4: Ablation Study of Query Size on the DCASE 2023 Task 2 Evaluation Dataset

Queries	Source AUC	Target AUC	pAUC	Official Score
1	81.60	76.55	67.27	74.65
4	82.22	77.27	66.67	74.81
8	82.28	76.75	68.42	75.38
16	83.30	79.40	68.32	76.46
24	84.02	80.45	69.83	77.62
32	81.99	81.99	66.01	74.03
64	83.48	76.11	67.06	74.94

Table 5: Ablation Study of Pooling Methods on the DCASE 2023 Task 2 Evaluation Dataset

Pooling Methods	Source AUC	Target AUC	pAUC	Official Score
Average pool	80.62	77.39	67.42	74.71
ASP [22]	82.91	77.11	67.17	75.15
FMQAP	84.02	80.45	69.83	77.62

der low FPR conditions. This indicates that FMQAP can learn the frequency patterns of different machines through attention mechanism, thereby strengthening the model’s generalization capability. While the F-WAP method gives only a limited performance boost, it substantially reduces the overall number of model parameters. Additionally, both L-WAP and data augmentation (spec-aug) contribute to slight performance gains.

In addition, we conducted experiments with different query sizes and various pooling methods, as shown in Table 4 and Table 5. The results in Table 4 indicate that performance gradually improves as the query size increases, reaching its peak at 24 queries. We attribute this to overfitting when the query size is too large, and insufficient learning of frequency patterns when the query size is too small. Meanwhile, Table 5 demonstrates that FMQAP significantly outperforms other frame-level pooling methods, yielding an official score improvement of 2.91% and 2.47%, respectively.

4. Conclusion

This paper has proposed a novel FMQAP method to finetune the pre-trained large audio model, *i.e.*, BEATs for addressing the first-shot problem in ASD tasks. FMQAP explicitly utilized the cross-band contextual information by multi-query attention pooling to improve the generalization capability for various machine types. The information extracted from different layers of pre-trained model is further exploited with L-WAP. Extensive evaluations have been conducted on DCASE 2020 and DCASE 2023 Task2 benchmarks. The results of 95.34% and 77.62% of mean score and official score have achieved, outperforming the previous STOA with a significant gap, demonstrating the effectiveness of the proposed FMQAP.

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